EE2004: End-Semester

May 9, 2025; Duration 3 Hours; Max of 50 marks

5 marks 1. State if the following statements are true or false and justify

(a) If the impulse response of a system is of finite duration, the system is stable.

(b) If continuous time signal has a bandwidth W (frequency response is limited to $[-W/2, W/2]$), then the sampling time T should be smaller than $1/W$ to prevent aliasing.

(c) If the input to a stable LTI filter $h[n]$ is periodic with period N, the output is periodic with period N.

(d) Every stable system (with a rational transfer function) has a stable inverse.

(e) Let $x[n]$, $n = 0, 1, ..L-1$ be a L length sequence. Suppose you sample the DTFT of the signal $x[n]$ at $\omega = 2\pi k/N$, $k = 0, 1, ..N-1$ for $N > L$ to obtain $X[k]$. If you now take an N point IDFT of the sequence $X[k]$, the signal $x[n]$ can be obtained from the output of the IDFT.

2. A real bandlimited periodic CT signal $x_c(t)$ has the exponential Fourier Series representation

$$x_c(t) = \sum_{k=-3}^{3} c_k \exp(jk\Omega_0 t),$$

where Ω_0 is the fundamental frequency in radians/sec.

2 marks (a) Determine the minimum sampling frequency Ω_s (to ensure no aliasing) for $x_c(t)$ in terms of Ω_0.

3 marks (b) Let $x_c(t)$ be sampled at $\Omega_s = 4.5\Omega_0$. Determine the frequencies of all the components in the resulting DT signal $\{x[n]\}$ in the range $[0, \pi]$.

3. Consider the N-point circular convolution (CC) of the sequences $h[\cdot] = \{h_0, \underset{\uparrow}{h_1}, h_2\}$ and $x[\cdot] = \{x_0, \underset{\uparrow}{x_1}, x_2, x_3, x_4\}$, where the $\uparrow$ indicates index zero and N is the length of $x[\cdot]$. Appropriate zero-padding is performed prior to CC. Let the output of the CC be the sequence $y[\cdot]$.

3 marks (a) Write an expression for *every* sample of $y[\cdot]$ in terms of samples of $x[\cdot]$ and $h[\cdot]$.

3 marks (b) Specify which samples of $y[\cdot]$ are identical to samples from the linear convolution of $x[\cdot]$ and $h[\cdot]$.

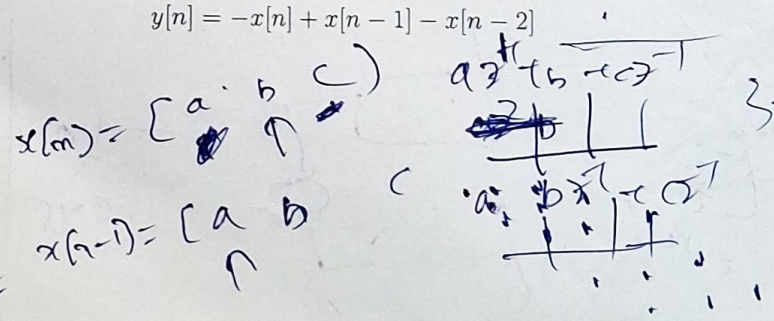
4. A filter has a frequency response given by $H(e^{j\omega})$. A new filter is defined whose frequency response is given by $G(e^{j\omega}) = H(e^{j(\omega-\pi)})$.

3 marks (a) if z_0 is a pole (zero) of $H(z)$, where is the corresponding pole (zero) of $G(z)$? (Justify)

3 marks (b) Express $g[n]$ in terms of $h[n]$.

5. Consider the LTI system S with input $x[n]$, output $y[n]$, and the difference equation:

$$y[n] = -x[n] + x[n-1] - x[n-2]$$



[1 marks] (a) What is the transfer function $H(z)$ for the system and the impulse response $h[n]$ of this system?

[1 marks] (b) Calculate the frequency response $H(e^{j\omega})$ of this system (DTFT of $h[n]$). Also determine the zeros of the frequency response, i.e., the values for ω where $H(e^{j\omega}) = 0$.

[1 marks] (c) Is this system stable? Does a stable inverse exist for this system? (Justify)

6. A sinusoid is sampled over one period to obtain the sequence $x[0] = 0, x[1] = 1, x[2] = 0$ and $x[3] = -1$. We want to do a 4-point DFT on this sequence using the DFT matrix F

$$\underbrace{\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix}}_{4\times 1} = \underbrace{[F]}_{4\times 4} \underbrace{\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}}_{4\times 1}$$

i.e., $\underline{X} = F\underline{x}$.

[2 marks] (a) Find the elements of matrix F.

[2 marks] (b) Using this matrix F find the DFT coefficients $X[0], X[1], X[2]$ and $X[3]$.

[3 marks] (c) If $Z[k] = \left[X[0]\ X[1]\ \underbrace{0\ 0\ \ldots\ 0\ 0}_{15}\ X[2]\ X[3]\right]$ then draw a sketch of $z[n] = \text{IDFT}\{Z[k]\}$. Please clearly indicate relevant point on the X and Y axis.

7. A stable system with transfer function $H(z)$ has a zero at $z = 4$, $z = 3$ and pole at $z = 1/2$ in the complex plane. It can be represented as the cascade of a stable minimum-phase system $H_{\min}(z)$ and a stable all-pass system $H_{\text{ap}}(z)$.

[4 marks] (a) Determine the choice of $H_{\min}(z)$ and $H_{\text{ap}}(z)$ (up to a scale factor).

[2 mark] (b) Draw the corresponding zero-pole plots.

[1 mark] (c) Indicate (and justify) if this decomposition is unique upto a scale factor.

8. Consider the following filter:

$$H(z) = \frac{1 + 0.95z^{-1}}{1 - 1.844z^{-1} + 0.849643z^{-2}}.$$

[2 mark] (a) Can you identify if it is a low-pass, high-pass or band pass filter? (Justify appropriately)

[2 marks] (b) Find the partial fraction decomposition of the above transfer function (useful information: $1.844 = 0.901 + 0.943$ and $0.849643 = 0.901 * 0.943$).

[3 marks] (c) As a practicing engineer, you are suppose to implement the following filter on a particular device. However because of the the limitations of the device, when implemented, the multiplier coefficients will automatically be quantized to the nearest multiples of 0.05. How will you implement this system on the device so that the final implemented filter is stable. Please provide the exact block diagram of the final filter realisation with clear labels.